GITAM Deemed to be University

Biotechnology | Course Code: BT301

Genetic Engineering & Recombinant DNA - Comprehensive Syllabus Lecture N

Syllabus Focus: BT301 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Genetic Engineering & Recombinant DNA. Includ

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Molecular Tools & Restriction Enzymes

" Syllabus Scope: Type II restriction endonucleases, DNA ligases, Alkaline phosphatases, Reverse transcri

Unit 2: Cloning Vectors & Transformation

" Syllabus Scope: Plasmids pBR322, pUC19, Bacteriophage lambda, Cosmids, YAC, BAC, Competent cell prepara

BT301: Genetic Engineering & Recombinant DNA - Advanced Topics & Assessment

Academic Track: Biotechnology - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Gene Amplification & Sequencing

" Polymerase Chain Reaction PCR variations RT-PCR, qPCR, Sanger chain termination sequencing, Next-Generation

Unit 4: Genome Editing & CRISPR-Cas9

" CRISPR-Cas9 target recognition, sgRNA design, Zinc Finger Nucleases, Recombinant therapeutics Insulin, Monoc

Unit 5: Expression Systems & Transgenic Organisms

" Prokaryotic vs Eukaryotic expression vectors, Inclusion bodies, Transgenic plants and animals, Bioethics and

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Describe the key structural components of pBR322 cloning vector and explain insertional inactivation f
2. Outline the thermal cycling phases of Polymerase Chain Reaction PCR and calculate theoretical amplicon
3. Explain the mechanism of CRISPR-Cas9 mediated double-strand break repair via NHEJ and HDR pathways.

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* T.A. Brown - Gene Cloning and DNA Analysis: An Introduction Wiley-Blackwell 7th Ed
* Bernard R. Glick - Molecular Biotechnology: Principles and Applications of Recombinant DNA ASM Press

Authorized NoteVault Educational Document " Verified for Registered University Students